

SCENARIO

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PROBLEM:

Help this person to manage his monthly spending on home loan and car loan by calculating both loans separately. You need to do a problem analysis and design (produce IPO Chart, Pseudocode and Flowchart) for the problems. Try to formulate the formula for both cases based on the sample interface attached.

SAMPLE OUTPUT:

Sample Output:

IPO CHART

INPUT	PROCESS	OUTP UT
Property_Price(RM)	Home_Loan_per_month=((Property_Price(RM)-Down_Payment(RM))*((Loan_Interest_Rate(%) / month_per_year)*(1+(Loan_Interest_Rate(%) / month_per_year)^(Housing_Loan_Period(Years)*month_per_year)))/((1+(Loan_Interest_Rate(%) / month_per_year)^(Housing_Loan_Period(Years)*month_per_year)-1)	Home_Loan_per_month
Down_Payment(RM)		
Housing_Loan_Period(Years)		
Loan_Interest_Rate(%)		

Variables:

Flout

Property_Price

;

Flout

Down_Paymen

t;

int

Housing_Loan

_Period;

Flout

Loan_Interest_

Rate;

PSEUDOCODE:

Start

Prompt user to enter Property_Price(RM)

Read Property_Price(RM)

Prompt user to enter Down_Payment(RM)

Read Down_Payment(RM)

Prompt user to enter Housing_Loan_Period(Years)

Read Housing_Loan_Period(Years)

Prompt user to enter Loan_Interest_Rate(%)

Read Loan_Interest_Rate(%)

$$\text{Home_Loan_per_month} = ((\text{Property_Price(RM)} - \text{Down_Payment(RM)}) * ((\text{Loan_Interest_Rate(\%)} / \text{month_per_year}) * (1 + (\text{Loan_Interest_Rate(\%)} / \text{month_per_year}))^{(\text{Housing_Loan_Period(Years)} * \text{month_per_year}))}) / ((1 + (\text{Loan_Interest_Rate(\%)} / \text{month_per_year}))^{(\text{Housing_Loan_Period(Years)} * \text{month_per_year})} - 1)$$

Home_Loan_per_month

End

FLOWCHART:

